"Education through self-help is our motto"
Rayat Shikshan Sanstha's
Yashavantrao Chavan Institute of Science, Satara
Constituent College of Karmaveer Bhaurao Patil University, Satara
NAAC GRADE 'A++' (CGPA 3.61)

Exam Seat No: **Date :**

CERTIFICATE

This is to certify, **Payal.M.Gaikwad (215) a**of Bachelor in Vocational Software Development have successfully completed the project,title **Student Attendance System.** during the academic year 2026-27.This report is submitted as partial report is submitted as partial fulfilment of the requirement of degree in B.voc Software Development of Karmaveer Bhaurao Patil University,Satara.

Teacher in charge Examiner HOD

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A

Project Synopsis
On

Student Attendance System

Submitted by

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YASHVANTRAO CHAVAN INSTITUTE OF SCIENCE
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Introduction:

The Student Attendance System is a java based application used to manage and maintain student attendance records. It uses JDBC and MySQL database to store, view, update and delete attendance information efficiently. The System helps reduce manual work and improve accuracy in attendance management.

Objectives:

- To maintain student attendance system record efficiently.
- To store attendance data in a MYSQL database.
- To add, view, update, and delete attendance records.
- To reduce manual paperwork and errors.
- To provide quick and easy attendance management using java and JDBC.

Scope Of The Project:

- The system is used to manage student attendance records digitally.
- It allows users to add ,view , update, and delete attendance information.
- Attendance data is stored securely in a MYSQL database.
- It reduces manual work and improve records accuracy.
- The project can be enhanced in the future by adding login ,repot generation,
- and attendance percentage calculation features.

The Existing System:

The existing attendance system is mostly manual, where attendance is recorded on paper registers or spreadsheets.

This process is time consuming, difficult to maintain, and prone to errors. Searching, updating, and managing attendance records manually can be inefficient, especially when handling a large number of student.

The Proposed System:

The proposed Student Attendance System is a computerized applicaton developed using Java, JDBC, and MySQL. It allows users to add, view, update, and delete student attendance records easily. The system stores data in a database, making attendance management faster, more accurate and efficient compared to the manual system.

Steps:

1. Study the existing manual attendance system.
2. Design the Student Attendance System using Java.
3. Connect Java with MySQL database using JDBC.
4. Create a table to store student attendance records.
5. Develop the Add Attendance module.
6. Develop the View Attendance module.
7. Develop the Update Attendance module.
8. Develop the Delete Attendance module.
9. Test all modules and verify database operations.
10. Display the final attendance records to the user.

Input Output:

Input:

```
import java.sql.*;
import java.util.Scanner;
public class StudentAttendanceDB {
Connection con;
StudentAttendanceDB() {
try {
Class.forName("com.mysql.cj.jdbc.Driver");
con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/attendance_db","root",""
);
System.out.println("Database Connected!");
} catch (Exception e) {
System.out.println(e);
}}
void addAttendance() {
try {
Scanner sc = new Scanner(System.in);
System.out.print("Enter Student ID: ");
int id = sc.nextInt();
sc.nextLine();
System.out.print("Enter Student Name: ");
String name = sc.nextLine();
System.out.print("Enter Attendance (Present/Absent): ");
String status = sc.nextLine();
PreparedStatement pst = con.prepareStatement("INSERT INTO attendance
VALUES(?,?,?)");
pst.setInt(1, id);
pst.setString(2, name);
pst.setString(3, status);
pst.executeUpdate();
System.out.println("Attendance Added!");
} catch (Exception e) {
System.out.println(e);
}}
void viewAttendance() {
try {
Statement st = con.createStatement();
ResultSet rs = st.executeQuery(
"SELECT * FROM attendance");
System.out.println("\nATTENDANCE LIST ");
while (rs.next()) {
System.out.println(
rs.getInt(1) + "  " +



rs.getString(2) + "  " +
```

```
rs.getString(3));
}
} catch (Exception e) {
System.out.println(e);
}}
void updateAttendance() {
try {
Scanner sc = new Scanner(System.in);
System.out.print("Enter Student ID to Update: ");
int id = sc.nextInt();
sc.nextLine();
System.out.print("Enter New Attendance (Present/Absent): ");
String status = sc.nextLine();
PreparedStatement pst = con.prepareStatement("UPDATE attendance SET status=?
WHERE id=?");
pst.setString(1, status);
pst.setInt(2, id);
pst.executeUpdate();
System.out.println("Attendance Updated!");
} catch (Exception e) {
System.out.println(e);
}}
void deleteAttendance() {
try {
Scanner sc = new Scanner(System.in);
System.out.print("Enter Student ID to Delete: ");
int id = sc.nextInt();
PreparedStatement pst = con.prepareStatement(
"DELETE FROM attendance WHERE id=?");
pst.setInt(1, id);
pst.executeUpdate();
System.out.println("Record Deleted!");
} catch (Exception e) {
System.out.println(e);
}}
public static void main(String[] args) {
StudentAttendanceDB s = new StudentAttendanceDB();
Scanner sc = new Scanner(System.in);
while (true) {
System.out.println("\nSTUDENT ATTENDANCE SYSTEM");
System.out.println("1. Add Attendance");
System.out.println("2. View Attendance");
System.out.println("3. Update Attendance");



System.out.println("4. Delete Attendance");
```

```
System.out.println("5. Exit");
System.out.print("Enter Choice: ");
int choice = sc.nextInt();
switch (choice) {
case 1:
s.addAttendance();
break;
case 2:
s.viewAttendance();
break;
case 3:
s.updateAttendance();
break;
case 4:
s.deleteAttendance();
break;
case 5:
System.out.println("Exit");
return;
default:System.out.println("Invalid Choice!");
}}}}
```

Output:

```
C:\Payal\JDBC>javac -cp ".;C:\Payal\JDBC\mysql-connector-j-8.0.33.jar" StudentAttendanceDB.java

C:\Payal\JDBC>java -cp ".;C:\Payal\JDBC\mysql-connector-j-8.0.33.jar" StudentAttendanceDB.java
Database Connected!
```

```
STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 1
Enter Student ID: 1
Enter Student Name: Payal
Enter Attendance (Present/Absent): Present
Attendance Added!

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 1
Enter Student ID: 2
Enter Student Name: Riya
Enter Attendance (Present/Absent): Present
Attendance Added!

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 1
Enter Student ID: 3
Enter Student Name: Vaishnavi
Enter Attendance (Present/Absent): Absent
Attendance Added!
```

```
STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 1
Enter Student ID: 4
Enter Student Name: Renuka
Enter Attendance (Present/Absent): Present
Attendance Added!

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 1
Enter Student ID: 5
Enter Student Name: Tanvi
Enter Attendance (Present/Absent): Absent
Attendance Added!

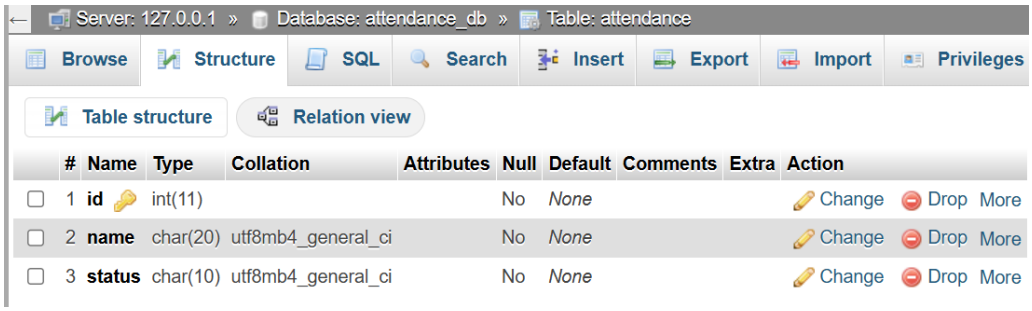
STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 2

ATTENDANCE LIST
1  Payal  Present
2  Riya  Present
3  Vaishnavi  Absent
4  Renuka  Present
5  Tanvi  Absent

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 3
Enter Student ID to Update: 2
Enter New Attendance (Present/Absent): Absent
Attendance Updated!

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 4
Enter Student ID to Delete: 4
Record Deleted!

STUDENT ATTENDANCE SYSTEM
1. Add Attendance
2. View Attendance
3. Update Attendance
4. Delete Attendance
5. Exit
Enter Choice: 5
Exit
```



Showing rows 0 - 3 (4 total, Query took 0.0004 seconds.)

```
SELECT * FROM `attendance`
```

id	name	status
1	Payal	Present
2	Riya	Absent
3	Vaishnavi	Absent
5	Tanvi	Absent

Software Requirement:

Software:

- Programming Language: Java(JDK 8 or above)
- MySQL Database
- JDBC Driver
- XAMPP(MySQL server)

Hardware:

- Processor: Intel i3 or higher.
- RAM: 4GB or more.
- Hard Disk: Minimum 500MB free space.

Future Scope of the Project:

- User Login and Authentication System.
- Attendance Percentage calculation.
- Automatically Report Generation.
- Student Search Facility.
- Web-Based or Mobile Application Integration.

Conclusion:

The Student Attendance System is a simple and effective application developed using java, JDBC, and MySQL. It helps in managing student attendance records efficiently, reduces manual work, and improves accuracy. The system providence an easy way to store, update, view, and delete attendance data.

Reference:

- Google Chrome